IPCON® PETG CF

Carbon Fiber Reinforced Poly (ethylene terephthalate-co-1,4-cyclohexylenedimethylene terephthalate) Filament / Carbon Fiber Reinforced Polyethylene terephthalate glycol Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	325 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.28 g/cm³
Shrinkage Rate*	Specially-defined	0.08% - 0.41%
Surface Hardness	Shore hardness	78D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	0.32%
Melt Index	250 °C, 2.16 kg	17.5 g/10 min
Melting Temperature	DSC, 10 °C/min	224 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	79 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	74 °C

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	47 MPa
Tensile Strength Z	ISO 527, GB/T 1040	31 MPa
Young's Modulus XY	ISO 527, GB/T 1040	3268 MPa
Young's Modulus Z	ISO 527, GB/T 1040	2248 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	15.4%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	14.7%
Flexural Strength XY	ISO 178, GB/T 9341	84 MPa
Flexural Strength Z	ISO 178, GB/T 9341	57 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	3488 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2376 MPa
Impact Strength XY	ISO 179, GB/T 1043	37.1 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	12.9 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 255 °C, bed temperature: 70 °C, printing speed: 100 mm/s) and IPCON PETG CF (Forged Iron Silver) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. High-grade metal look, antique look; higher strength and stiffness, excellent toughness and layer adhesion strength; less warping, higher precision; water and oil resistance, upgraded weather resistance, long-term outdoor use; easy to print, compatible with AMS and CFS, supports high speed (max speed of more than 250 mm/s).
2. Trendy toys, cultural and creative works, art ornaments, electronic DIY product casings, FPV drone frames, remote control car casings, potted plant containers, etc.

• Recommended Printing Settings

Drying Settings Before Printing	65 - 70 °C, 6 - 8 h (Blast drying oven)
Printing Temperature and Humidity	≤ 45 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PETG
Compatible Printer Type	Open-frame, enclosed-frame (Suggest to open the front door and remove the upper lad)
AMS or CFS	Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.4 mm / 0.6 mm / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick
Nozzle Temperature	245 - 270 °C
Bed Temperature	65 - 75 °C
Fan Speed	10% - 90%
Max Print Speed	250 mm/s

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